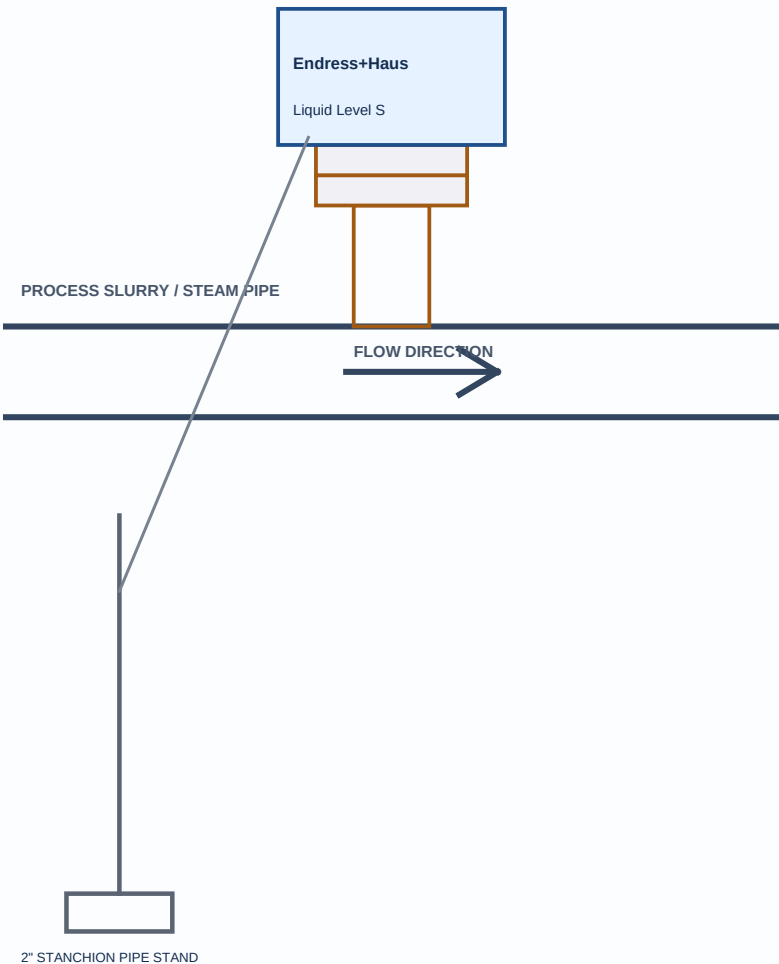


TEMPLATE TYP-12: LIQUID LEVEL SWITCH

TYPICAL TAGS: LSH-40101 / LSL-40101

SERVICE: High / Low Slurry Level Interlock & Pump Protection | PLC I/O: 1756-IB32 (24V DC Sinking/Sourcing Digital Input)

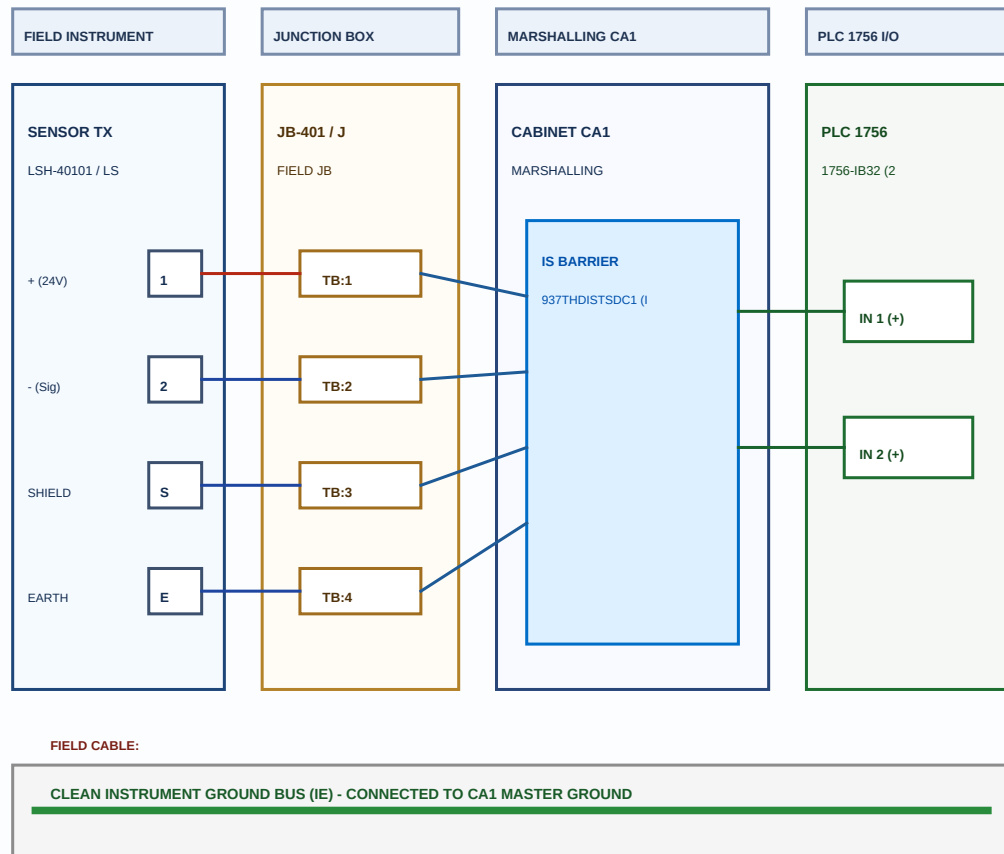
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	3/4" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Stainless ..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6.....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Level Alarm (Dry / Submerged ..
Power Supply:	24V DC (10-55V DC, PNP Mode)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THDISTSDC1 (IS Digital Input Barrier)
PLC Module:	1756-IB32 (24V DC Sinking/Sourcing Dig..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Liquiphant FTL51B (Tuning Fork SS316L)

SERVICE: High / Low Slurry Level Interlock & Pump Protection

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Liquiphant FTL51B Level Switch E+H Tuning Fork SS316L (100mm)		1 EA
2	Welding Boss 3/4" NPT Half Cou SS316 Tank Welding Socket		1 EA
3	Cable Gland M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		1 EA
4	Viton / EPDM Thread Seal Ring Process Sealing Gasket		1 EA
5	Grounding Strap 4mm² with Lugs Copper Flexible Earth Braid		1 EA

ENGINEERING INSTALLATION NOTES

- Install fork with tines aligned vertically to prevent slurry buildup on top of vibrating blades.
- Set fail-safe mode switch: MAX (Overfill Protection) or MIN (Dry Run Pump Protection).
- Keep fork at least 50mm away from tank walls, internal baffles, and agitator blades.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTLIQUID LEVEL SWITCH (VIBRATING FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-012
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
14 / 25